Machine Learning – CDS201

Module 2: Regression Models

Multiple Linear Regression

- The multiple linear regression explains the relationship between one continuous dependent variable (y) and two or more independent variables (x1, x2, x3… etc).
- By adding more predictors to the simple linear regression model, this technique helps to understand better how the predictors affect the outcome variable as a whole.
- Using an equation that best fits the observed data, the main objective of multiple linear regression (MLR) is to forecast the dependent variable's value based on the independent variables' values.

Assume that there are multiple independent variables *say* x_1, x_2, ...x_n. If the relationship between independent variables x and dependent or output variable y is modeled by the relation,

$$y=a_0+a_1*x_1+a_2*x_2+.....+a_n*x_n$$

Multiple Linear Regression – solved example

- For the dataset,

x1 Product 1 Sales	x2 Product 2 Sales	Y Weekly Sales
1	4	1
2	5	6
3	8	8
4	2	12

- We need to calculate a, where a_0, a_1, a_2 are the intercept, and coefficient of x1 and x2

$$a = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}.$$

- Regression coefficient

$$\hat{a} = ((X^T X)^{-1} X^T) Y$$

Multiple Linear Regression – solved example

$$X^T X = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}$$

$$(X^T X)^{-1} = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}^{-1} = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix}$$

$$X^T X)^{-1} X^T = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix} \times \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 4 & 5 & 8 & 2 \end{pmatrix} = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.098 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix}$$

$$\hat{a} = ((X^T X)^{-1} X^T) Y = \begin{pmatrix} 0.05 & 0.47 & -1.02 & 0.19 \\ -0.32 & -0.098 & 0.155 & 0.26 \\ -0.065 & 0.005 & 0.185 & -0.125 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix} = \begin{pmatrix} -1.69 \\ 3.48 \\ -0.05 \end{pmatrix}$$

$$\boldsymbol{a_0 = -1.69}$$
$$\boldsymbol{a_1 = 3.48}$$
$$\boldsymbol{a_2 = -0.05}$$

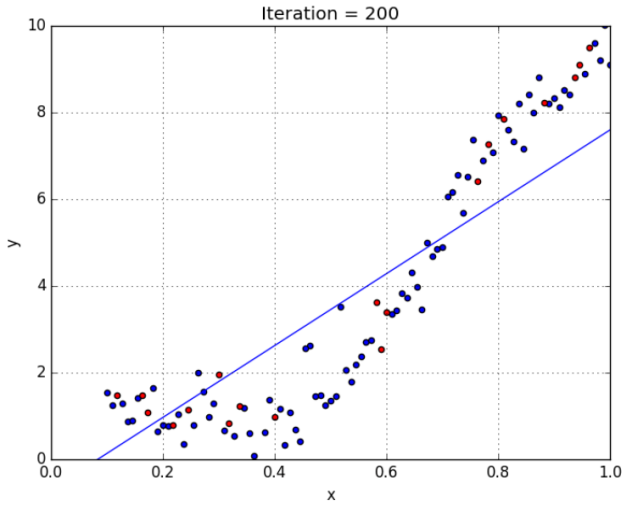
Multiple Linear Regression – solved example

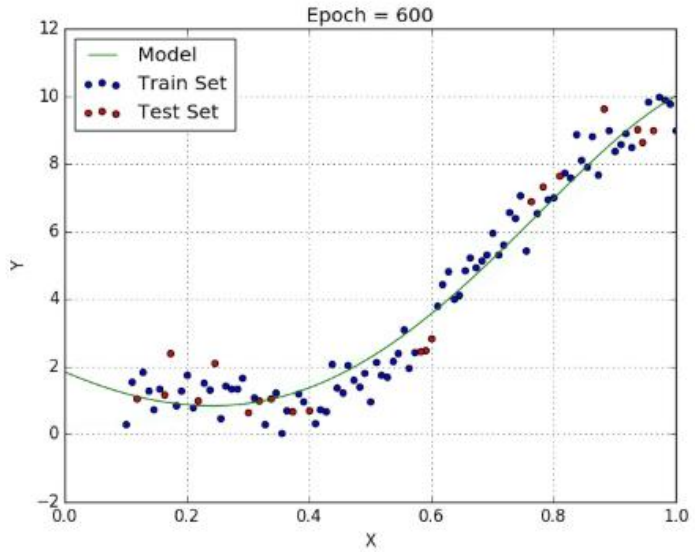
- Multiple linear regression model

$$y = -1.69 + 3.48x_1 - 0.05x_2$$

- Knowing the values of x1 and x2 , you can predict for any ‘y’.

Polynomial Regression





Polynomial Regression is a statistical technique that models the relationship between a dependent variable y and an independent variable x as an nth degree polynomial. It's an extension of linear regression, used when the data shows a non-linear relationship.

$$\boxed{y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_n x^n + \epsilon}$$

- y is the dependent variable.
- x is the independent variable.
- $\beta_0, \beta_1, \ldots, \beta_n$ are the coefficients of the model.
- ϵ is the error term.

- Determining the degree of the polynomial:

  - Use cross-validation techniques to evaluate how well models of different degrees generalize to unseen data. K-fold cross-validation is commonly used, where the data is split into K subsets. The model is trained on K-1 of these subsets and validated on the remaining subset, with the process repeated K times so that each subset is used as the validation set once. The degree that results in the lowest average validation error is typically chosen.

  - Generally, polynomials of maximum degree 4 are used, as even higher order polynomials take some strange shapes and overfit the data. (Low bias and high variance)

Polynomial Regression – solved example

- Find a quadratic regression model for the following data:

X	Y
3	2.5
4	3.2
5	3.8
6	6.5
7	11.5

Let the quadratic polynomial regression model be

$$y=a_0+a_1*x+a_2*x^2$$

- Regression parameters

$$\boldsymbol{a} = \boldsymbol{X}^{-1}\boldsymbol{B}$$

$$X = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \sum y_i \\ \sum(x_i, y_i) \\ \sum(x_i^2, y_i) \end{bmatrix}$$

	x	y	x^2	x^3	x^4	y*x	$y*x^2$
	3	2.5	9	27	81	7.5	22.5
	4	3.2	16	64	256	12.8	51.2
	5	3.8	25	125	625	19	95
	6	6.5	36	216	1296	39	234
	7	12	49	343	2401	80.5	563.5
Σ	**25**	**27.5**	**135**	**775**	**4659**	**158.8**	**966.2**

$a_0 = 12.4285714$

$a_1 = -5.5128571$

$a_2 = 0.7642857$

- The required quadratic polynomial model is

$$y = 12.4285714 - 5.5128571 * x + 0.7642857 * x^2$$

Regularization

- Regularization is employed to adjust the model to fit the test set effectively and properly.
- Regularization techniques are crucial in minimizing overfitting and ensuring the model performs optimally.
- Regularization in machine learning serves as a method to forestall a model from overfitting. Overfitting transpires when a model not only discerns the inherent pattern within the training data but also incorporates the noise, potentially leading to subpar performance on fresh, unobserved data.
- The employment of regularization aids in mitigating this issue by augmenting a penalty to the loss function employed for model training.

- When completely new data from the problem domain is fed as an input into deep learning models, regularization is a collection of strategies that can assist prevent overfitting in neural networks and improve their accuracy by modifying the learning procedure slightly such that the model generalizes more successfully.
- The model then performs better on the unobserved data as a result.

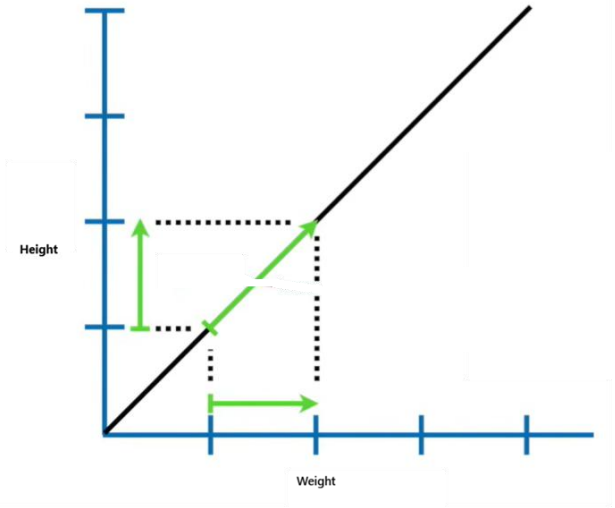
- Regularization is implemented by modifying the loss function. In a typical machine learning model, the loss function measures how well the model's predictions match the actual data. Regularization adds an extra term to this loss function that penalizes large weights (slope).

$$Loss = Original\ Loss + \lambda \times Regularization\ Term$$

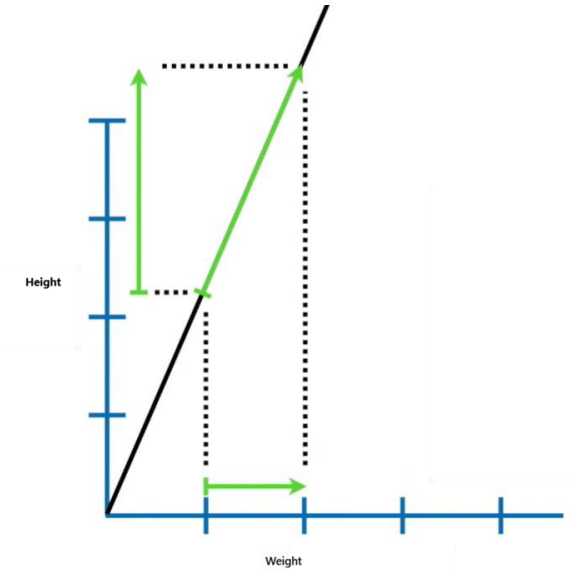
- **Original Loss**: Measures the difference between the model's predictions and the actual values.
- **Regularization Term**: Adds a penalty for larger weights.
- **λ (lambda)**: A hyperparameter that controls the strength of the penalty. A larger λ means more regularization.

- By minimizing this new loss function, the model not only fits the data but also keeps the weights small, which helps in generalizing better to new data.

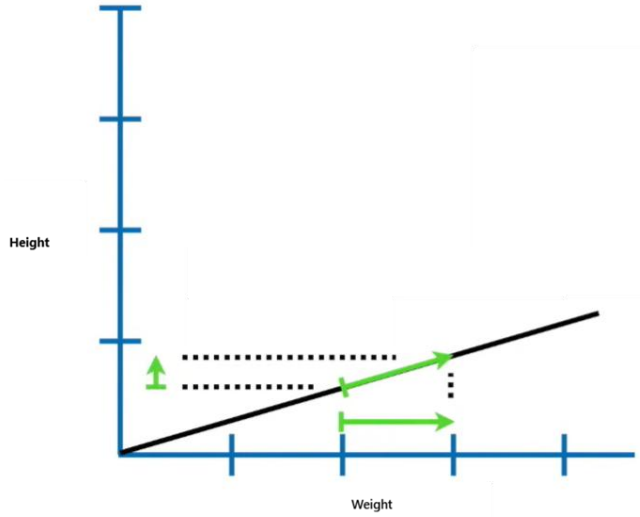
Effect of the Regularization term



When the slope is 1 there is one unit change in weight it is captured by 1 unit change in height



When the slope is high the prediction for height is very sensitive to relatively small changes in weight



When the slope is low the prediction for height is less sensitive to relatively changes in weight

- The choice of λ (also known as the regularization parameter) is crucial. It is typically chosen via cross-validation to balance fitting the training data well and keeping the model simple enough to perform well on new data.

- There are several types of regularization techniques commonly used:
  - L1 Regularization (Lasso): This adds a penalty equal to the absolute value of the magnitude of coefficients. This can lead to some coefficients being zero, which means the model ignores the corresponding features. It is useful for feature selection.
  - L2 Regularization (Ridge): Adds a penalty equal to the square of the magnitude of coefficients. All coefficients are shrunk by the same factor, and none are eliminated, as in L1.
  - Elastic Net: This combination of L1 and L2 regularization controls the model by adding penalties from both L1 and L2, which can be a useful middle ground.

- **L1 Regularization (Lasso Regression)**
  - It adds the absolute value of the weights to the loss function.

$$Loss = Original\ Loss + \lambda \sum_i |W_i|$$

  - Imagine we have a dataset with many features (variables), but not all of them are important for predicting the output.
  - Using L1 regularization can help the model focus on the most significant features by reducing the weights of less important ones to zero.

- **L2 Regularization (Ridge Regression)**
  - It adds the square of the weights to the loss function.

$$Loss = Original\ Loss + \lambda \sum_i W_i^2$$

  - Suppose we're building a model to predict house prices based on various features like size, number of rooms, age, location, etc.
  - L2 regularization helps ensure that the model doesn't assign too much importance to any one feature and considers all of them in a balanced way.

- **Elastic Net:**
  - The Elastic Net is a regularized regression technique combining ridge and lasso's regularization terms.
  - The *r* parameter controls the combination ratio. When *r=1*, the L2 term will be eliminated, and when *r=0*, the L1 term will be removed.

$$\text{Elastic Net Cost Function} = \text{Loss Function} + \lambda \left(r \sum_{j=1}^{m} |w_j| + \frac{(1-r)}{2} \sum_{j=1}^{m} w_j^2 \right)$$

  - Although combining the penalties of lasso and ridge usually works better than only using one of the regularization techniques, adjusting two parameters, λ *and r*, is a little tricky.